

अर्थ AI: End-to-End AI-Augmented Architecture for Financial Auditing and Report Generation

Capstone Project Proposal

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Mentor Consent Form

I hereby agree to be the mentor of the following Capstone Project Team:

Project Title: अर्थAI: End-to-End AI-Augmented Architecture for Financial Auditing and Report Generation

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Project Overview

The financial audit process in India is one of the most document-intensive, regulation-heavy, and legally consequential workflows that a small business is required to undergo every year — yet it remains almost entirely manual. अर्थAI is built to change that. The foundation of the system is a curated dataset assembled from publicly available company filings sourced from the Ministry of Corporate Affairs portal, augmented with synthetically generated financial records designed to cover edge cases and anomaly patterns not sufficiently represented in public data. This raw financial data is processed and normalised into json format. Real audit reports corresponding to these financials form the ground truth, establishing a supervised training objective: given raw financial data, the system must learn to produce a correctly structured, appropriately worded audit report without altering a single numerical value present in the source material.

Once the core report generation capability is established and validated against ground truth, a dedicated anomaly detection model is trained on the same structured financial data to identify patterns that warrant auditor attention — irregular transaction sequences, statistical outliers, ratio breaches, and entries inconsistent with the broader financial picture. This module operates under a strict mathematical correctness constraint: every flag it raises must be traceable to a specific rule or regulation or it should be a precedent to previous cases.

The final layer of the system is a Retrieval-Augmented Generation pipeline constructed over the applicable regulatory and legal corpus governing statutory audits in India. Regulations are stored as a queryable index and retrieved at inference time in response to specific financial observations and anomaly flags. The three components — structured data extraction, anomaly detection, and regulatory RAG — are brought together in a human-reviewed report generation step, where the auditor sees the AI's draft, the evidence behind it, and the regulation it cites, and approves or corrects each section before the report is finalised. The result is a system where AI handles the volume and the structure, and the auditor handles the judgement.

Need Analysis

The necessity for an AI-augmented audit assistance system is grounded in a structural mismatch between the scale of India's corporate compliance obligations and the technological resources available to the practitioners responsible for meeting them. As of January 2025, a total of 28,05,354 companies are registered in India, of which approximately 65 per cent — over 18 lakh entities — are active and conducting business. Every one of these active companies is legally required to undergo a statutory audit annually, regardless of turnover, profit, or business activity, with non-compliance attracting fines and restrictions on fund-raising and borrowings. This creates an audit demand of enormous scale that falls overwhelmingly upon small and medium CA practitioners, who handle the majority of engagements outside the large corporate sector.

The audit process itself is operationally intensive. For small CA firms handling ten to fifteen clients simultaneously, this manual workload is compounded by the expanded reporting obligations. No currently available tool assists the auditor in substantively populating given clauses from underlying financial data.

Global firms such as Deloitte, PwC, and KPMG have developed proprietary AI-assisted audit tools — Omnia, GL.ai, and Clara respectively — but these are closed, internally deployed systems unavailable to independent CA firms. Academic and commercial research confirms the gap: the iRAA platform, one of the few publicly reported LLM-integrated audit systems, achieves accuracy rates of only 70–80 per cent without human oversight, underscoring that current AI tools are not yet safe for autonomous deployment and must be paired with structured human review. SusGen-GPT [7] demonstrates that domain-adapted models in the 7–8B parameter range can approach GPT-4-level performance on financial NLP tasks, establishing that high-quality AI assistance is achievable within the compute constraints of on-device deployment. The system proposed in this project directly addresses all three layers of this gap — the operational burden on small audit firms, the absence of end-to-end regulatory tooling, and the need for human-in-the-loop safeguards — making it both timely and practically significant.

Literature Survey

1. Large Language Models for Audit and Financial Report Generation

Wang et al. (2025) [6] present one of the most rigorous attempts at automating the financial statement audit process end-to-end using LLMs. The system is given a company's historical transaction records alongside a financial statement that may or may not contain errors, and it is asked to judge whether the statement is correct, identify the specific error, explain why it is wrong, cite the accounting standard that is violated, and finally produce a corrected version of the statement. While GPT-4 correctly identifies whether a statement is erroneous in nearly all cases, its ability to pinpoint the exact error type falls to around 75% accuracy for single errors and degrades further to around 59% for multiple errors. More critically, its overall success rate — meaning it gets every single stage right — is only 4.1% for single errors and 3% for multiple errors. The authors specifically identify the absence of domain-specific accounting knowledge and the difficulty of reasoning jointly across tabular data and unstructured text as the two root causes — findings that directly motivate the design decisions in the present project.

Xu et al. (2025) [8] approach the same problem from a different angle. Rather than testing whether a general-purpose LLM can audit out of the box, a BERT-based encoder-decoder architecture is trained specifically for audit report generation and compliance classification. The encoder produces contextual representations of input audit texts, and a Transformer decoder is trained to generate report text that conforms to professional audit standards, while a parallel classification head assigns compliance risk labels to each text segment. Trained and evaluated on public financial audit datasets spanning multiple industries, the system achieves a BLEU score of 0.78 and a compliance classification accuracy of 92.5%, outperforming GPT, T5, and LSTM baselines across all metrics. However, robustness testing reveals a significant weakness — when the input text contains noise or tampered data, accuracy drops to 78% and the F1-score falls to 76%. The authors acknowledge that their system is built entirely on clean, pre-digitised text and has not been tested on scanned or multimodal financial documents, and that it does not adapt to dynamic regulatory changes. Both of these limitations are directly addressed in the proposed system.

2. Dataset Construction for Financial NLP

Wang et al. [6] describe real transaction data from companies as "hardly applicable" due to confidentiality, so accounting students are engaged to write few-shot examples that are then used to prompt GPT-4 to generate synthetic transaction data, followed by a round of human verification. Financial statements are sourced from S&P 500 companies, captured as screenshots and transformed into structured text. Errors are then systematically injected into these statements — missing rows, numerical errors, redundant rows, and misclassifications — to create a labelled benchmark dataset of 371 correct statements and 1,855 erroneous ones.

Wu et al. (2024) [7] demonstrate a more elaborate version of the same principle at scale. The SusGen-30K dataset is assembled from publicly available annual reports and open-source financial datasets through a five-stage automated pipeline: collection, translation of multilingual content into English, reformatting into an instruction-following format, anonymisation of company-identifying information using LLMs, and augmentation of under-represented task categories through LLM-assisted synthesis. The result is a 30,000-sample, category-balanced dataset spanning seven financial NLP tasks. The ablation study is particularly instructive — model performance improves consistently and measurably as dataset size increases from 10k to 20k to 30k samples across every task, confirming that dataset scale and quality are the primary drivers of performance in this domain rather than model architecture alone.

3. Domain Adaptation and On-Device Efficiency of Smaller Models

A practical constraint that separates real-world audit tooling from research prototypes is compute. An audit assistance system that requires a cloud API call for every inference is a system that cannot be used — client financial data cannot leave the client's premises, and small CA firms do not have enterprise cloud budgets. The question of whether a small, locally deployable model can match the quality of a large frontier model on financial tasks is therefore not academic — it is the central feasibility question for this entire project.

Wu et al. [7] answer this question convincingly. SusGen-GPT is a suite of 7-8B-parameter models fine-tuned using QLoRA — a parameter-efficient method that quantises the model to 4-bit precision and trains only a small set of additional low-rank adapter weights, enabling fine-tuning on two consumer-grade NVIDIA RTX 3090 GPUs. Despite being more than 200 times smaller than GPT-4 by parameter count, the best SusGen-GPT model achieves an average financial benchmark score of 0.55 against GPT-4's 0.57 — a gap of only 0.02. On several specific tasks including relation extraction and financial table question answering, it actually surpasses GPT-4. This is not a marginal result — it is a direct demonstration that domain-adapted small models, trained on carefully curated financial data, can deliver frontier-level quality at a fraction of the compute cost. The present project adopts the same strategy and the same 7-8B parameter model range for exactly this reason.

4. Retrieval-Augmented Generation for Regulatory Grounding

The single most dangerous failure mode in an audit assistance system is a model confidently citing a regulation that does not say what the model claims it says. Wang et al. [6] document this failure empirically — even GPT-4 achieves only a 26.2% Top-1 accuracy on standard citation, meaning it correctly identifies the relevant accounting standard in fewer than one in four cases when working from its parametric knowledge alone. This is not a model quality problem — it is a fundamental architectural problem. Regulatory knowledge embedded in model weights cannot be updated when rules change, cannot be verified, and cannot be cited with a source link. The solution is to not embed regulatory knowledge in the model at all.

Wu et al. [7] demonstrate the alternative architecture through the RAG component of the SusGen system for sustainability report generation. Rather than asking the model to recall TCFD standards from memory, relevant passages are retrieved at inference time from a vector-indexed corpus of annual reports using chunk-level semantic embeddings, with the top-10 most relevant chunks combined into the prompt before generation. The generated report is therefore grounded in retrieved text rather than in model weights, and every claim in the output can be traced back to a specific retrieved passage.

5. The India-Specific Gap and the Motivation for This Project

All three reviewed works, despite their complementary contributions, share a common limitation that **defines the space this project occupies**: they are built entirely on Western financial systems. Wang et al. [6] use S&P 500 financial statements governed by FASB standards. Wu et al. [7] use TCFD-format sustainability reports from multinational corporations. Xu et al. [8] evaluate on multi-industry datasets with no Indian regulatory context.

The gap is therefore not incremental — it is structural. A system trained on FASB-governed balance sheets cannot reason about CARO clause obligations. A RAG pipeline indexed on TCFD standards cannot answer questions about related-party loan thresholds under Section 185 of the Companies Act. A report generation model trained on S&P 500 data will not produce outputs in the format expected by ICAI. The present project is the first attempt to build an end-to-end AI-augmented audit assistance pipeline specifically designed for the Indian statutory audit context, combining the methodological contributions of the reviewed works — synthetic dataset construction, and RAG-grounded report generation — with the regulatory specificity and human-in-the-loop design that Indian audit practice legally requires.

Objectives

The following four objectives are to be achieved at the conclusion of the capstone project:

1. **Objective 1: Securely read and extract data**

Safely process uploaded company documents—including scanned images and PDFs—and automatically pull the essential text and numbers from them.

2. **Objective 2: Organize the financials**

Take that raw, extracted data and automatically assemble it into clear, standardized balance sheets and financial summaries.

3. **Objective 3: Spot issues and quote the rules**

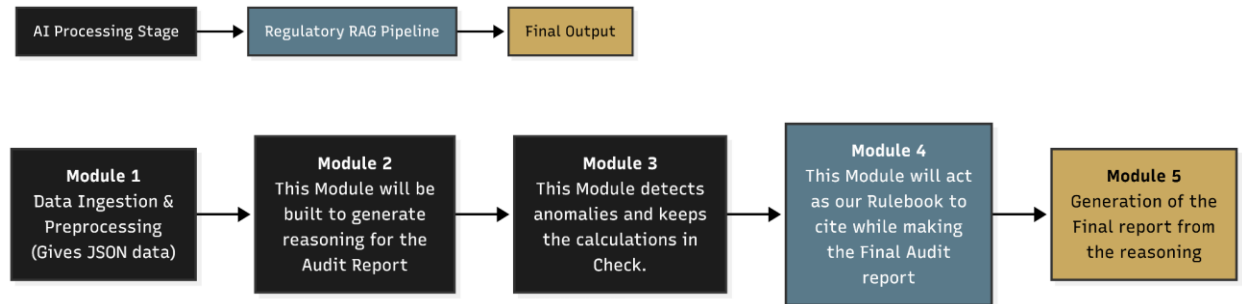
Review the financial transactions to flag unusual patterns or potential risks, and automatically pull up the specific accounting rules that govern those exact items.

4. **Objective 4: Draft the report (with human oversight)**

Write a complete, evidence-based draft of the audit report. The system will give the human auditor full control to easily review, edit, approve, or reject any section before the final version is created.

Methodology

A 5 module approach is utilized in making this system for End-to-End AI assisted auditing.



Flowchart showing the different modules in the pipeline

1. **Module 1: Data Ingestion & Preprocessing:** Raw data from various sources (databases, spreadsheets, financial records) is ingested, cleaned, and standardized into a structured **JSON format**. This ensures every downstream module receives consistent, machine-readable input without ambiguity or formatting issues.
2. **Module 2: Reasoning Generation:** The structured JSON is fed into an LLM which analyzes the data and **generates human-like reasoning and observations** — identifying patterns, summarizing key figures, and forming initial audit narratives. This is the "thinking brain" of the pipeline.
3. **Module 3: Anomaly Detection & Calculation Validation:** Before anything reaches the final report, this module acts as a **quality control gate**. It cross-checks numerical calculations, detects statistical outliers, and flags any suspicious entries that deviate from expected norms — keeping the audit honest and accurate.
4. **Module 4: Rulebook / Compliance Reference:** This module serves as the **legal and regulatory backbone** of the pipeline. When the audit report makes a claim or flags an issue, it cites the relevant rule, standard, or policy (e.g., GAAP, IFRS, SOX) from this module — making every conclusion defensible and auditable.
5. **Module 5: Final Report Generation:** In the last section we generate the final report that is editable for the auditor.

Work Plan

Table representing the monthly plan for the project

Sr. No.	Activity	Month	March	April	May	June	July	August
1	Literature Review	Plan						
		Actual						
2	Dataset collection and synthetic augmentation	Plan						
		Actual						
3	Architecture Design and Pre-processing	Plan						
		Actual						
4	Develop Module 2 for audit report reasoning	Plan						
		Actual						
5	Develop Module 3 for anomaly detections	Plan						
		Actual						
6	Develop Module 4 RAG Pipeline	Plan						
		Actual						
7	Testing and Validation	Plan						
		Actual						
8	Final Presentation	Plan						
		Actual						

Project Outcomes & Individual Roles

Outcomes:

1. **Outcome 1:** An end-to-end, AI-powered audit platform that transforms raw financial data into government rule-compliant audit outputs, with full traceability, intelligent anomaly detection, and auditor-in-the-loop validation.
2. **Outcome 2:** A research paper based on the findings and methodologies used in the project, as well as benchmarking the performance of different language models in such a specific task.

Roles:

Prakhar Raj (102317004) - Deep Learning & Backend Development

Saumya Gupta (102317028) - Regulatory RAG and NLP

Bhavya Kalal (102317076) - Frontend Development

Sumit (102315019) - AI/ML Integration and documentation

Tanjot (102303411) - Conversational systems and documentation.

Course Subjects

The following course subjects and their associated conceptual knowledge and practical skill sets are directly applied in the execution of this capstone project:

- Machine Learning (UML501) — Supervised and unsupervised learning algorithms, neural network architectures, model evaluation metrics; applied in the anomaly detection and journal entry classification modules.
- Natural Language Processing (UCS664) — Text representation, transformer architectures, retrieval-augmented generation, semantic similarity; applied in the regulatory RAG pipeline and clause response generation.
- Database Management Systems (UCS310) — Relational schema design, query optimisation; applied in structuring the canonical ledger schema and the audit evidence store.
- Computer Networks and Web Technologies — Client-server architecture, REST APIs; applied in the human-in-the-loop review interface and pipeline service communication.
- Software Engineering (UCS503) — Requirements analysis, system design, version control, testing methodologies; applied throughout the project for structured development and documentation.
- Operating Systems (UCS303) — Process management, file I/O; applied in the on-device inference runtime and document processing pipeline.
- Probability and Statistics (UMA401) — Statistical distributions, hypothesis testing, Benford's Law; applied in the rule-based anomaly detection component.
- Data Structures and Algorithms (UCS301) — Applied in efficient parsing, indexing, and retrieval operations across the pipeline.

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